Predictive Maintenance – Compressed Air System

STANDARD OPERATING PROCEDURE (SOP)

Purpose

This Standard Operating Procedure (SOP) defines the mandatory process for responding to predictive maintenance alerts related to Industrial Compressed Air Systems (COMP_1, COMP_2) used for primary facility pressure and pneumatic control. The objective is to ensure safe, consistent interventions to prevent catastrophic pressure drops, undetected manifold leaks, and premature compressor motor failure.

Scope and Applicability

This SOP applies to all high-pressure pneumatic systems and compressor units located in the Pneumatic Control Room. It must be followed by authorized maintenance personnel whenever predictive interventions are triggered by the AI Engine and delivered through the Smart Action Pack via the M365 Copilot interface.

Roles and Responsibilities

- **Maintenance Technician** - Responsible for executing all physical inspections, pneumatic leak diagnostics, and corrective maintenance steps in this SOP.
- **Site Supervisor** - Responsible for verifying task completion, validating system pressure stabilization, and approving work order closure.
- **Reliability Engineer** - Responsible for monitoring post-intervention compressor duty cycles, adjusting predictive alert thresholds, and refining the AI logic.

Overview

This SOP details the mandatory actions required when a predictive maintenance alert indicates a sustained drop in system air pressure or an abnormally high compressor duty cycle. It outlines the safe procedures for isolating the pneumatic lines, identifying high-pressure leaks, clearing blockages, and restoring normal operating pressure.

Trigger Conditions (Entry Criteria)

This SOP is initiated when all the following conditions are met:

- Main pneumatic line pressure drops below 6.5 bar for a sustained period exceeding 10 minutes.
- The compressor motor duty cycle exceeds 85% unexpectedly, indicating it is overworking to maintain pressure.
- A Smart Action Pack notification is issued by the AI Engine.

Health & Safety Rules

- Standard site Personal Protective Equipment (PPE) must be worn: safety glasses with side shields, steel-toe boots, and impact-resistant gloves.
- **Mandatory Hearing Protection:** Ambient noise levels in the Pneumatic Control Room frequently exceed 85 dB; heavy-duty ear protection is required.
- **High-Pressure Hazard:** Never use bare hands or body parts to feel for air leaks. High-pressure air can cause severe injection injuries.
- Systems must be fully depressurized and safely vented before uncoupling any hoses, replacing filters, or opening valves.

Required Tools

- Mobile device with M365 Copilot and D365 applications installed.
- Ultrasonic acoustic leak detector (for identifying micro-leaks in high-noise environments).
- Calibrated digital pressure gauge for manual verification.
- Pneumatic Lockout/Tagout (LOTO) kit including valve lockouts and bleed-off tags.
- Replacement coalescing filters and standard adjustable wrenches.

Step-by-Step Execution Process

- **1. Preparation:** Acknowledge the dispatch from the Smart Action Pack and review the AI-generated pressure loss timeline and anomaly summary on your mobile device.
- **2. Locating the Asset:** Locate COMP_1 inside the Pneumatic Control Room using the asset routing details.
- **3. Initial Inspection:** Conduct an initial scan of the compressor housing and main discharge manifold using the ultrasonic acoustic leak detector to identify any immediate high-pressure air leaks.
- **4. Power & Pneumatic Isolation:** Apply LOTO protocols to the compressor motor's electrical disconnect. Close the primary high-pressure discharge isolation valve to separate the compressor from the facility air loop.

- **5. Depressurization:** Slowly open the localized bleed valve to safely vent all residual compressed air trapped between the compressor and the isolation valve. Verify zero pressure on the local analog gauge.
- **6. Component Check & Correction:** Inspect the main coalescing filter housing; replace the filter element if heavily fouled with oil/particulates (a common cause of sudden pressure drops). Tighten or replace any manifold couplings flagged by the ultrasonic detector. Check the pressure relief valve (PRV) for signs of premature venting.
- **7. System Restart:** Close the bleed valve, reseal all filter housings, remove the LOTO equipment, and re-energize the compressor. Once the compressor builds initial pressure, slowly open the primary discharge isolation valve to repressurize the facility loop.
- **8. Observation:** Monitor the localized telemetry dashboard for 10 minutes. Ensure the pneumatic system pressure reaches and holds a steady setpoint of approximately 7.5 bar, and verify the compressor duty cycle returns to normal limits (typically 40-60%).

Exit Criteria

This SOP may be considered complete only when all of the following conditions are met:

- System operating pressure has stabilized at approximately 7.5 bar.
- Compressor duty cycle is verified to be within normal baseline limits.
- No pneumatic leaks are detected during a post-repair ultrasonic sweep.
- The system has operated continuously for a minimum 10-minute observation period without fault.

Exception Handling and Escalation

If system pressure continues to drop or the compressor motor fails to normalize its duty cycle:

- Do not leave the system running continuously.
- Re-apply pneumatic and electrical lockout conditions.
- Escalate immediately to the Site Supervisor and the Reliability Engineering team for advanced diagnostics (e.g., internal rotary screw inspection or facility-wide pipe leak detection).

Post-Job Audit and Verification

The following evidence is mandatory and must be uploaded to the D365 work order record:

- Results of the pre- and post-intervention ultrasonic leak detector sweeps.
- Photographic evidence of the replaced filter elements (if applicable).
- Confirmation of pressure stabilization and duty cycle normalization.

All checklist items must be digitally signed off within the M365 Copilot interface. Work orders must not be closed until all evidence is verified and approved by the Site Supervisor.

SOP Governance

SOP Owner:	Reliability Engineering Team
Version:	1.0
Approval Authority:	Head of Maintenance
Effective Date:	05/05/2026
Review Cycle:	Annual or upon system change